

The Feather Report: *Trochilidae* and Female-Limited Polymorphism

Sexual dimorphism is best described as the morphological differences between sexes in the same species; these variations are not necessarily associated with interior morphology or physical function but usually occur as differences in size or appearance between the sexes (Darwin 1871, cited in Diamant et al. 2021). The most obvious example is the plumage variation in birds – in that case, the evolution and prevalence is typically influenced by sexual selection pressures (Andersson 1994, cited in Diamant et al. 2021). This phenomenon manifests as colorful plumage and appearance for males, while females of the same species are often cryptic as a result of safety selection (Andersson 1994, cited in Diamant et al. 2021). Polymorphs are common across several bird species; the hummingbird clade *Trochilidae* is no exception. They are often used to study sex-linked dimorphism due to males being under stronger sexually selection; however, there are instances of female individuals exhibiting the colorful, “catchy” appearance associated with males. Eleanor Diamant and Dustin Rubenstein of Columbia University with Jay Falk of UCLA sought to explore this female-limited polymorphism. Female-limited polymorphism, or FLP going forward, is a condition in which a species has a single male form (andromorphs) and multiple female forms (heteromorphs) (Diamant et al. 2021); in some species, the dual morphology is linked to sexual selection, particularly in polygynous species in which males have multiple female mates (Fitzpatrick et al. 2018, cited in Diamant et al. 2021). This study utilized the perceived prevalence of FLP across several species of *Trochilidae* to examine why this morphology might be selected for in females of a polygynous species.

By examining this phenomenon, it invites the possibility that non-sexual factors may be at work in the selective pressures faced by a species as morphological variation develops; also, the morphological variation may be an adaptive response to selection (i.e., it serves a purpose) or non-adaptive (random and not likely to be sustained). The authors established multiple hypotheses addressing either circumstance: possible adaptive hypotheses for this evolution of FLP include social selection, sexual selection, and / or safety selection (Dale, et al. 2015, LeBas 2006, Alonzo 2008, cited in Diamant et al. 2021). Androchromic presentation might protect a female from male harassment, may result from sexual selection in mate selection, may indicate competitive social selection between the same or opposite sex, or the androchromic morph (male mimic) may have better social success or resource access (Tobias et al. 2012, cited in Diamant et al. 2021). Non-adaptive hypotheses posit that FLP is an intermediary state due to strong male selection, due to pleiotropy (a single gene influences multiple phenotypic traits), or selection on other traits leading to polymorphic variation (linked to andromorphic bill length here) (Bonduriansky et al. 2009, Iserbyt et al. 2013, cited in Diamant et al. 2021). These hypotheses are grounded in concepts and data that already exist; in some species, this FLP can result from competition and interaction within the sexes - social selection influences the evolution of female polymorphs more so than mate selection (LeBas 2006, cited in Diamant et

al. 2021). Both adaptive and non-adaptive hypotheses were plugged into existing hummingbird data; resource competition is linked to sexual dimorphism and bill length, and those individuals often exhibit behavioral dimorphism as well (more aggression in males), so the androchrome polymorph may also operate as a “social cue”.

In addition to pulling from previous data, researchers acquired 300 hummingbird skins from multiple species of the family *Trochilidae* from various museums; sexed and labeled skins were already available. The study established qualitative and quantitative values for the prevalence and features to identify FLP (Del Hoyo et al. 1999, cited in Diamant et al. 2021). Researchers also created a statistical model of evolution to identify ancestral states and determine the evolutionary origin and number of times the trait of FLP has developed. To begin testing the non-adaptive hypotheses, researchers put together “ancestral character states and transition rates” (Diamant et al. 2021); they then compared the rates of transition from sexual monochromatism (having only one color) to sexual dichromatism (involving coloration between sexes) to rates of transition between those states reversed (Lande 1980, McGuire et al. 2007, cited in Diamant et al. 2021). If transition rates from monochromatism to dichromatism were higher, then FLP is non-adaptive; females have a different optimal phenotype for the same trait, resulting in conflicting selection pressures. To test the pleiotropy hypothesis, researchers looked at andromorphic bill length and assessed whether that particular phenotype was linked to the androchromic plumage. Color photographs of male and female hummingbirds from 22 species were analyzed, and a researcher measured the extent of coloration and bill length (Rasband 2015, cited in Diamant et al. 2021). The authors acknowledge exclusion of measuring degree of ultraviolet light reflectance, despite its previously established correlation with plumage vibrance (Falk 2019, unpublished data, cited in Diamant et al. 2021) and the possibility that birds may see variation in plumage that is imperceptible to the human eye (Burns et al. 2012, cited in Diamant et al. 2021). Researchers used social-linked traits that already had data available for each species: whether they are migratory (IUCN 2016, cited in Diamant et al. 2021), social hegemony (Bribiesca et al. 2019, cited by Diamant et al. 2021), and body size. Ecological aspects considered were mean temperature and precipitation, and temperature and precipitation predictability (Harris et al. 2020, Colwell 1974, cited in Diamant et al. 2021). Using phylogenetic comparative methods to test the adaptive hypothesis, researchers generated a statistical model of analysis using degree of FLP as the dependent variable, and migratory status, social influence, and body size the independent variables (Orme et al. 2018, cited in Diamant et al. 2021). Some limiting factors to the overall analysis include that only sexually dichromic (having sex-linked coloration) species were included in the analysis, and there was a risk of mis-sexing a specimen. Researchers opted to take a conservative approach; since this study relies on knowing the specific sex of the polymorph, any samples for which sex could not be wholly verified were excluded from evaluation. In contrast to the author’s self-assessment of “conservative” FLP sexing, Christopher Clark, Barbara Robinson, and James Remsen question Diamant et al.’s determination of the prevalence of FLP; they felt that their own research showed FLP to be more rare and required a more dimensional and conscientious analysis, and the previous study likely mis-sexed specimens by attempting to identify the presence of testes or ovaries, which is considerably difficult given the hummingbird’s size (Clark et al, 2022).

Results of the author's study suggest that approximately a quarter of known species showed evidence of FLP and indicated that the trait has independently evolved 28 times across the clade. Results did not indicate that FLP is an intermediate state between monochromatism and dichromatism. Researchers found no pleiotropic relationship between androchrome coloration and bill length, although some other traits were associated with each other (Clark et al. 2020, cited in Diamant et al. 2021). Both non-adaptive hypotheses were rejected; however, researchers did link socioecological factors to FLP (the one factor without any association was body size). The most significant relationship was to migratory status. Migratory species were less likely to exhibit FLP (Gass 1979, Rousseau et al. 2020, cited by Diamant et al. 2021). To a lesser degree, socially dominant species trended toward exhibiting FLP more so than nondominant species. There was also a significant relationship with the ecological factor of lower mean precipitation. These social and ecological factors have been associated with competition over resources with members of different species sharing the same ecological niche, suggesting that the incidence and frequency of FLP in hummingbirds is an adaptive social selection (Hixon et al. 1983, Marquez-Luna et al. 2019, cited in Diamant et al. 2021). More investigation is necessary to identify the precise relationship between the socioecological factors and evolution of FLP (Is it a status symbol? Do we contextualize it by intersexual competition?), although this tentative association between them will support further research in exploring the evolution of sexual dimorphism as a result of socioecological selective pressures (Kunte 2009, cited by Diamant et al. 2021).

Naturally there are big limitations to these results; by studying behavioral data and skins separately, it omits the possibility of observing behavior and appearance working in tandem as a function of social selection. A possible future for this kind of research is tagging, sexing, and observing live specimens across multiple generations, ecological niches, and species; while it would take a lot of time and patience, it would possibly yield more insightful data and indicate the extent to which physical, temporal, and spatial circumstances may be influential. The authors themselves indicate that these findings are only preliminary and should be part of a more extensive effort to investigate the extent and direction of statistical correlation of socioecological factors to the presence and persistence of FLP, to explore why factors like migratory status and climate factors have a relationship to the evolution and persistence of FLP, and to consider other related factors in this assessment (Bent 1940, Murray 1981, Rousseau et al. 2020, Dalsgaard et al. 2011, cited in Diamant et al. 2021). Criticisms of this study include noting the omission of key factors that should have been considered in assessing the true prevalence of FLP, such as age of each individual and the time of specimen acquisition (noting that mis-sexing is likelier to occur in older specimens), along with the possibility that Diamant et al. may have over-reported the prevalence of FLP in the limited species that were studied (Clark et al. 2022). Newer research actually indicates that FLP is more rare in hummingbirds than previously assessed (Clark et al. 2022). Despite these limitations and criticisms, it opens the door to further study and highlights the need for revisiting biological study and statistical correlations, not just for the "how" and "why" of organisms studied, but to reassess the methodology, bias, and construct of research.

In any case, morphological mimicry and variation between individuals in a species offers vital insight into a species communication, selection, and survival across all Animalia

(Rubenstein et al. 2009, Dijkstra et al. 2009, Vercken et al. 2007, cited in Diamant et al. 2021); this study in particular felt important, and bigger than its flaws, as its core message seeks to understand sexual dimorphism beyond the confines of sexual selection. The newer study that highlights the limitations and flaws in the author's study also contributes to the discourse; the criticism seeks to improve the methodology in order to further the same understanding of FLP sought by the author. These types of evolutionary studies are vital to refining human knowledge of biological processes; that knowledge has been historically contextualized by the framework of a long-standing patriarchy (which thankfully is losing its chokehold on many dimensions of society), and its pervasiveness undoubtedly influences our subconscious perceptions, biases, assumptions, and hypotheses about other species' behaviors. As we learn more about sex-linked traits in animals, it is necessary to recontextualize and reexamine our notions of sex-linked traits and behavior in the larger context of environment, climate, species-specific behaviors, ecological niche, and essentially expand beyond traditional assumptions that may be guided by a very unintentional social or cultural human bias subconsciously "gendering" sex-linked morphology and appearance. Plus, it is essential to maintain the integrity of the scientific method to set no limitation on the extent to which anything can be studied – there is always something to learn, whether or not anyone is looking for answers!

Literature Cited:

Clark, C. J., Robinson, B., & Remsen, J. V. (2022). Female plumage polymorphism is rare in hummingbirds. *Journal of Ornithology*, 1-14.

Diamant, E. S., Falk, J. J., & Rubenstein, D. R. (2021). Male-like female morphs in hummingbird the evolution of a widespread sex-limited plumage polymorphism. *Proceedings of the Royal Society B*, 288(1945), 20203004.

*I changed my citing article to one that felt more relevant as a contrast to the study conclusions from the original article (as it is a direct response to the original study)